Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **scalable distributed systems**, **performance-tuned backends**, and simulation and benchmarking pipelines in **C++ and Python**, with strong database, concurrency, and latency-optimization foundations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Database Systems, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed **highly scalable distributed backend systems** with fault-tolerant async orchestration across 5K+ monthly jobs, improving throughput **3x** under concurrent load while sustaining 99%+ availability.
- Reduced **latency from 1.2s to 350ms** for 500+ users by profiling hot paths and tuning SQL access patterns, indexing, and storage design across multi-tenant database systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **concurrent processing engine** with multithreading, benchmarking and tuning throughput across 1K+ files per batch and 10K+ daily requests for stable performance under sustained load.
- Built **scalable backend services and instrumentation** tracking per-operation cost and usage, driving data-driven efficiency and reliability improvements across high-volume production workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and benchmarking pipelines** over 100K+ records for closed-loop evaluation, enabling reproducible performance correlation and high-confidence debugging of distributed model-driven systems.
- Profiled and tuned a **performance-critical model** through rigorous benchmarking and experimentation, improving long-horizon evaluation **18%** across large-scale simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **distributed data pipelines and APIs** (AWS Glue, S3) with multi-region build and test automation, improving response times **30%** and pipeline efficiency **68%** across cross-platform systems.
- Built **VM-based simulation and test infrastructure** (nightly builds, 20+ suites) with Jenkins and PyTest across 3 OSes, accelerating debugging and increasing reliability for distributed releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **distributed microservices** with WebSocket real-time sync across eight workflows, scaling to 2M+ records on Azure SQL while optimizing throughput and low-latency responsiveness.
- Automated **Docker-based deployment and monitoring** via Jenkins with Azure Monitor, cutting release time **50%** and improving availability and stability across production environments.

Skills

Languages: C++, Python, Java, SQL, JavaScript/TypeScript, Bash
Distributed Systems: Scalability, High Availability, Fault Tolerance, Concurrency, Multithreading, Async Orchestration
Performance: Benchmarking, Profiling, Latency Optimization, Performance Tuning, Debugging, Load Testing
Data & Storage: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Data Modeling, Indexing, Query Optimization
Simulation & Testing: Simulation Frameworks, Test Automation, Unit/Integration Testing, Closed-Loop Evaluation
Cloud & Tooling: AWS (S3, Glue, ECS, RDS), Azure, Docker, Kubernetes, CI/CD, Jenkins, Git, Linux

Projects

CPU-Scheduling Simulator Live | *C++, Simulation, OS Internals*
- Built a **C++ simulator** modeling FCFS, SJF, SRT, Round Robin, and multi-level queue scheduling algorithms with Gantt visualization, reducing average turnaround **25%** across 1K+ simulated process configurations.

Chat2SQL Code | *Data Storage Systems, Java, TypeScript*
- Built a workspace integrating **10+ relational and NoSQL data storage systems** (PostgreSQL, MySQL, MongoDB, Redis) with schema-aware querying and Docker-backed cross-platform deployment in Java and TypeScript.

Big-Brain Live | *Distributed Data Platform, Performance Tuning*
- Built a **distributed cloud data platform** indexing 50K+ records with incremental updates and performance tuning, cutting retrieval latency **40%** through optimized execution paths.